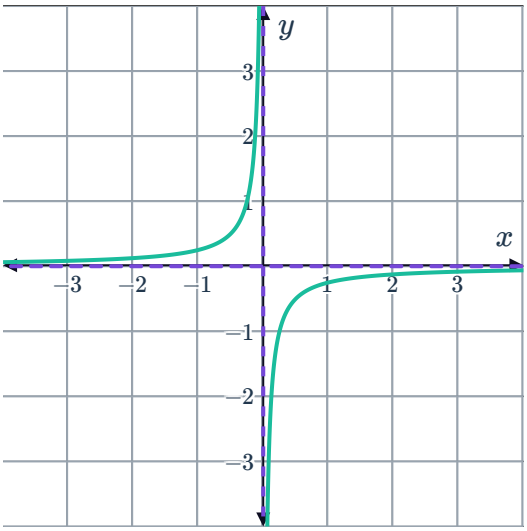
Graphs of transformed hyperbolas

1

a

x	-3	-2	-1	1	2	3
y	$\frac{1}{12}$	$\frac{1}{8}$	$\frac{1}{4}$	$-\frac{1}{4}$	$-\frac{1}{8}$	$-\frac{1}{12}$

b



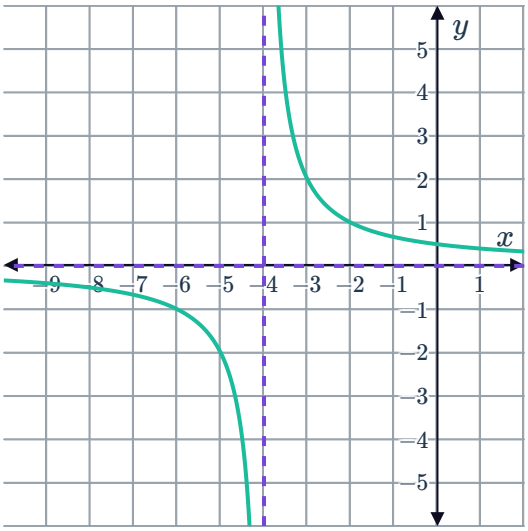
c 2 and 4

2

a $x = -4$

b $y = 0$

c



3

a

i No

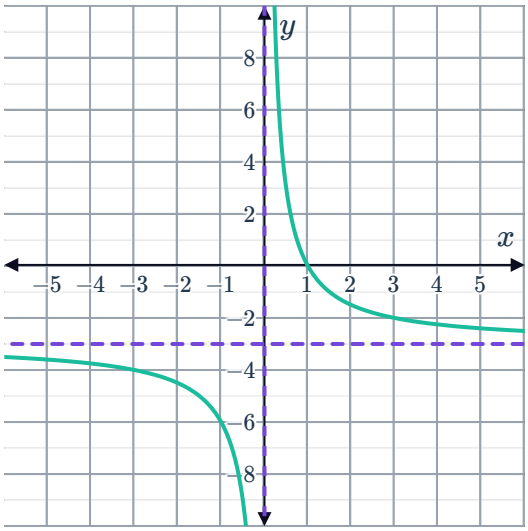
ii Yes

b $y = \frac{3}{x} - 3$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c $x = 0, y = -3$

d



4

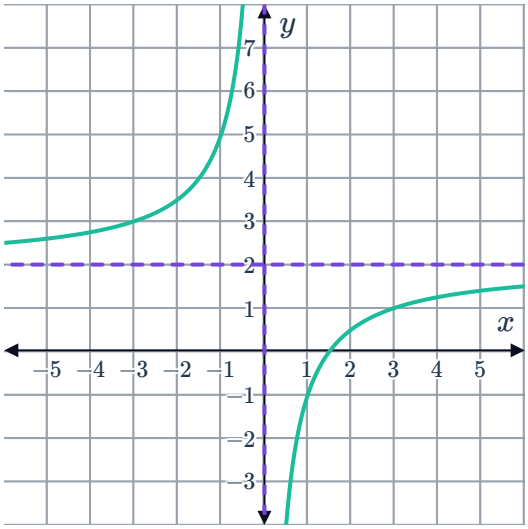
a

i Yes **ii** No

b $y = \dfrac{-3}{x} + 2$

c $x = 0, y = 2$

d



5

a

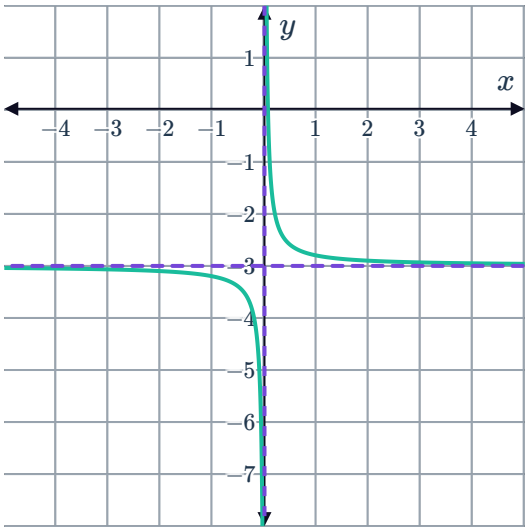
i Yes **ii** No

b $y = \dfrac{1}{x}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c $x = 0, y = -3$

d



6

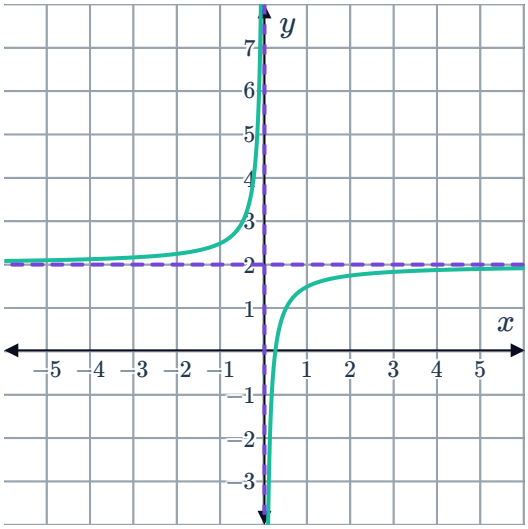
a

i Yes **ii** No

b $y = \dfrac{-1}{x}$

c $x = 0, y = 2$

d

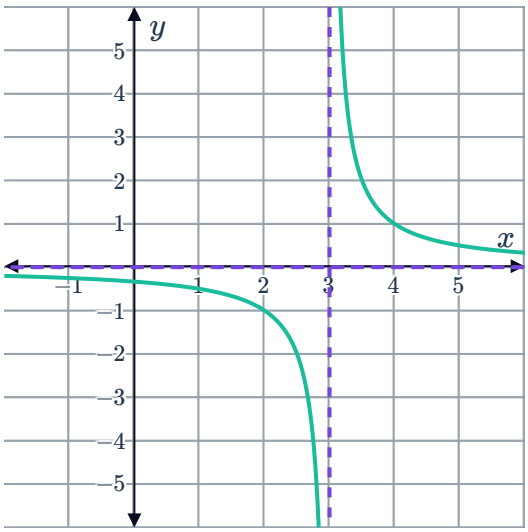


7

a

x	1	2	$\frac{5}{2}$	$\frac{7}{2}$	4	5
y	$-\frac{1}{2}$	-1	-2	2	1	$\frac{1}{2}$

b



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

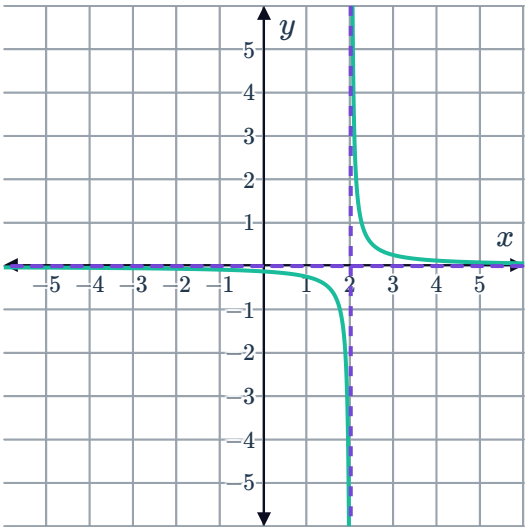
8

a

i **B**

ii $x = 2, y = 0$

iii

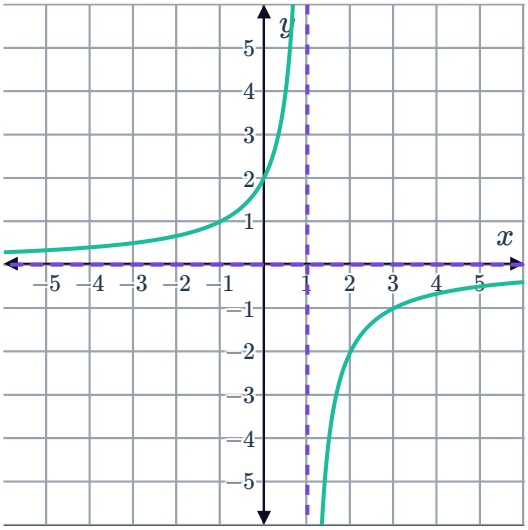


b

i **A**

ii $x = 1, y = 0$

iii



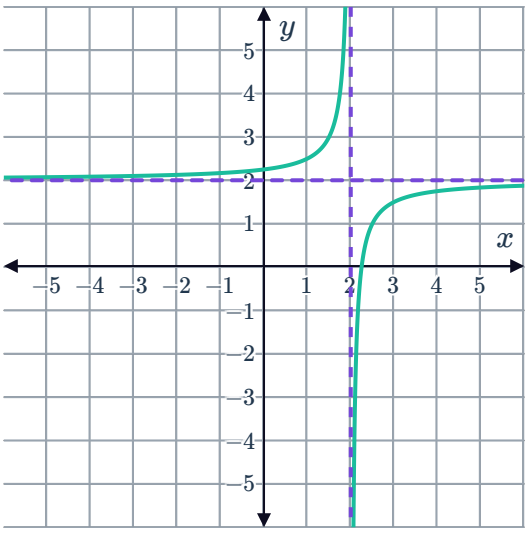
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c

i **A**

ii $x = 2, y = 2$

iii



9

a 2

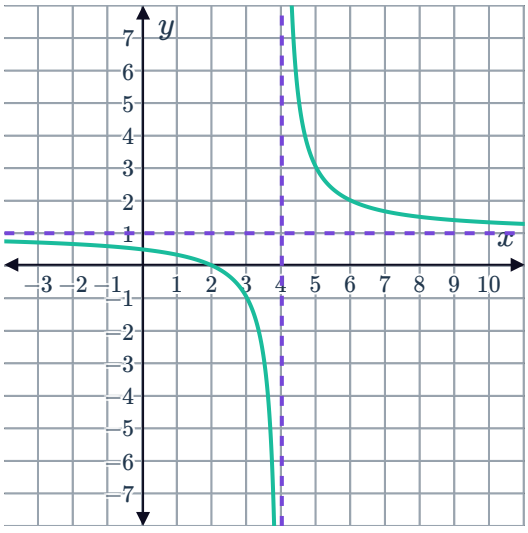
b $y = \dfrac{2}{x-4} + 1$

c $x = 4$

d 1

e $y = 1$

f



10

a 1

b $y = -\dfrac{1}{x-3} + 1$

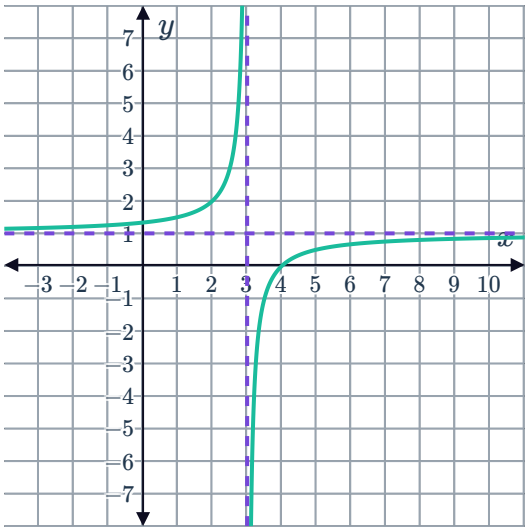
c $x = 3$

d 1

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

e $y = 1$

f



11

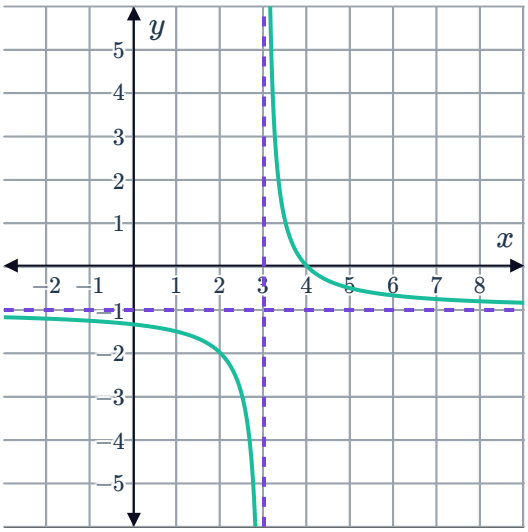
a $y = \dfrac{1}{x-3} - 1$

b $x = 3$

c -1

d $y = -1$

e



Transformations and domain and range

12

a $y = \dfrac{1}{x} + 4$

b $y = \dfrac{1}{x-7}$

13

a -3

b -6

c Translated 3 units down

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14

a $x = 0$

b 2 and 4

c $x = 4$

d $1, 2$ and 4

e Translated 4 units right

15

a Yes

b No

c Yes

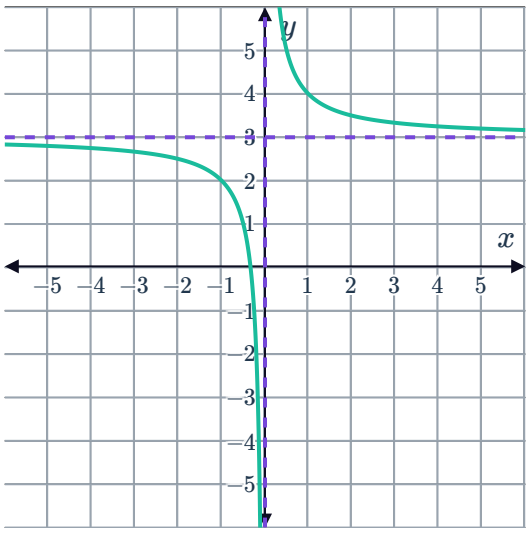
d No

16

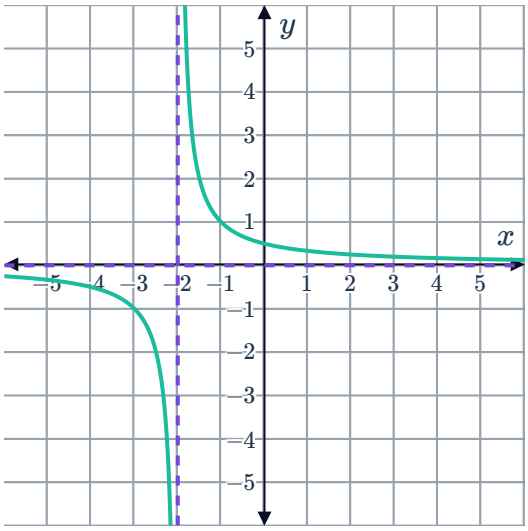
a Move the graph upwards by 3 units.

b Shift the graph to the left by 2 units.

c



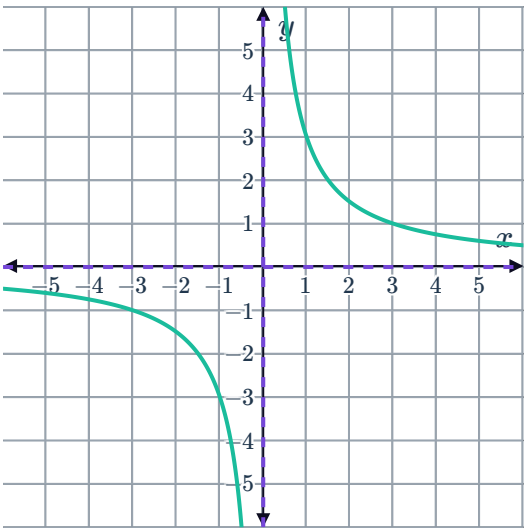
d



17

a Expansion away from the asymptotes by a factor of 3.

b



c $(-\infty, 0) \cup (0, \infty)$

d $(-\infty, 0) \cup (0, \infty)$

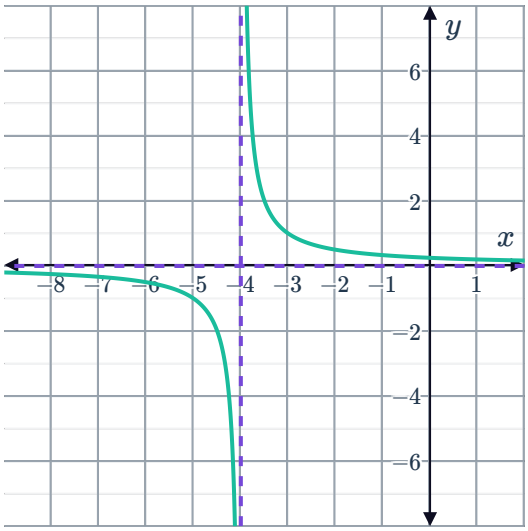
e Decreasing

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18

a Shift to the left by 4 units.

b



c $(-\infty, -4) \cup (-4, \infty)$

d $(-\infty, 0) \cup (0, \infty)$

e Decreasing

19

a $(-\infty, -1) \cup (-1, \infty)$

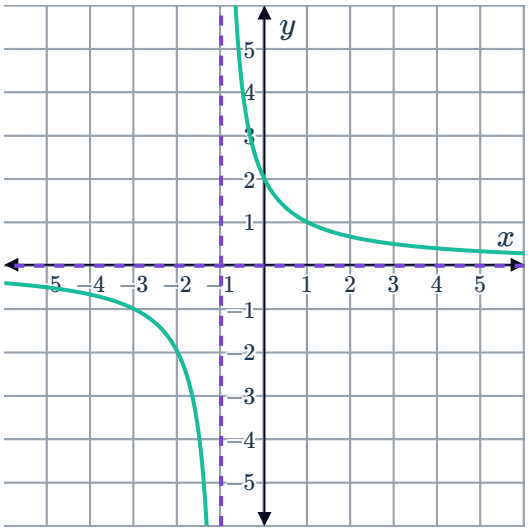
b $x = -1$

c $x = \dfrac{2}{y} - 1$

d $(-\infty, 0) \cup (0, \infty)$

e $y = 0$

f



20

a $x \in \mathbb{R}, x \neq 0$

b $x = 0$

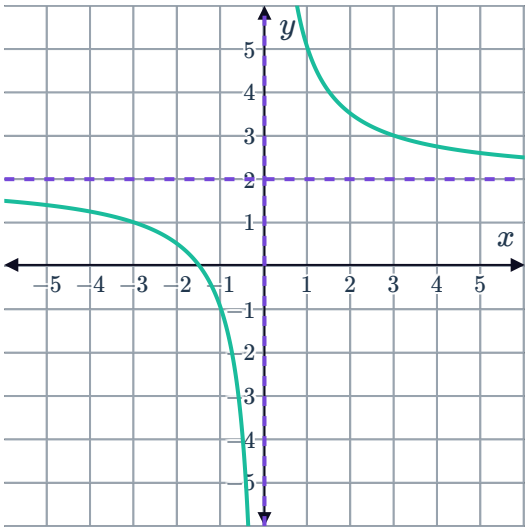
c $x = \dfrac{3}{y - 2}$

d All real $y \neq 2$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

e $y = 2$

f



21

a All real $x \neq 1$

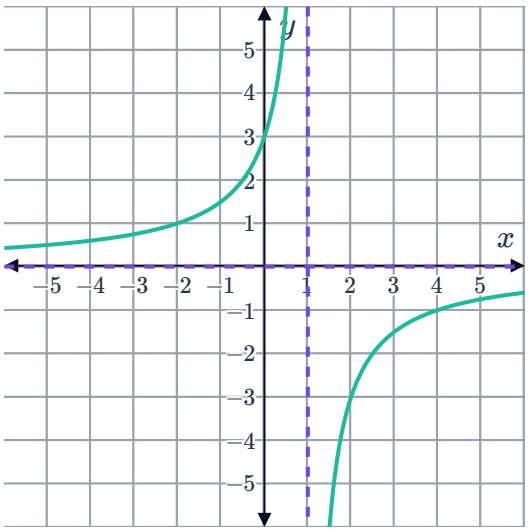
b $x = 1$

c $x = 1 - \dfrac{3}{y}$

d All real $y \neq 0$

e $y = 0$

f



22

a $a = -2$

b $x = -2$

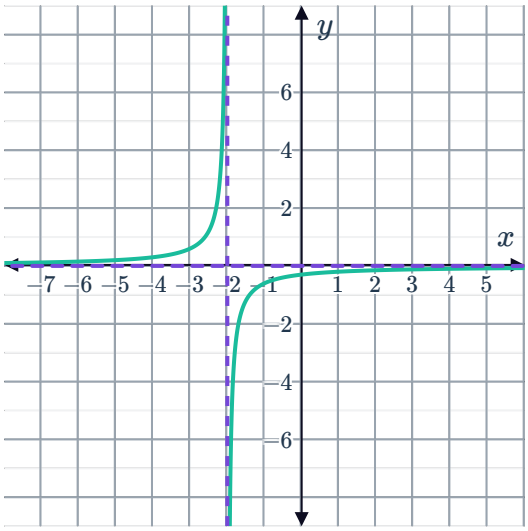
c 0

d $y = 0$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

e 0

f



Applications

23

a $k = 1220$

b $t = 40$

24

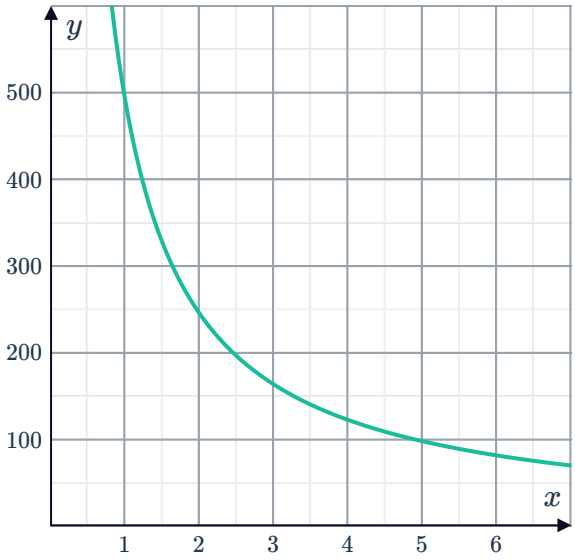
a $\$245.00$

b $y = \dfrac{490}{x}$

c

x	1	2	3	4	5	6
y	490	245	163.33	122.5	98	81.67

d



e Inverse Variation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

$\frac{15}{2}$ units

© ClipSAT: By Mr. Mohamed Abdallah +966597688647